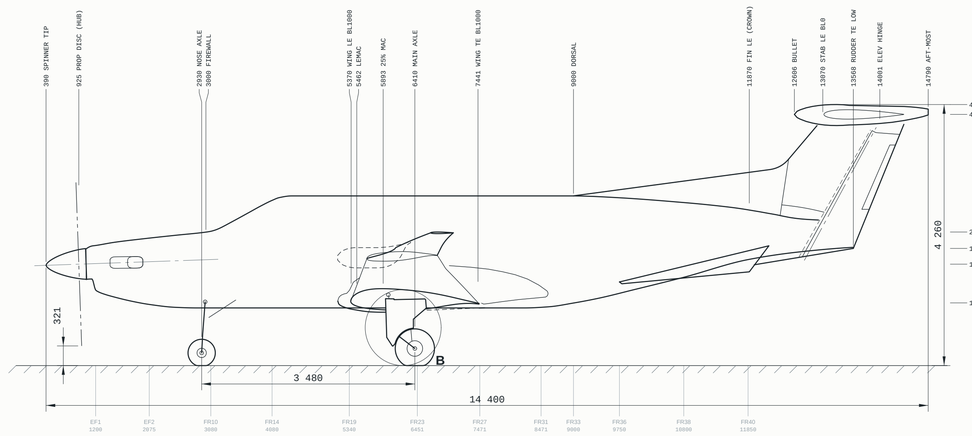
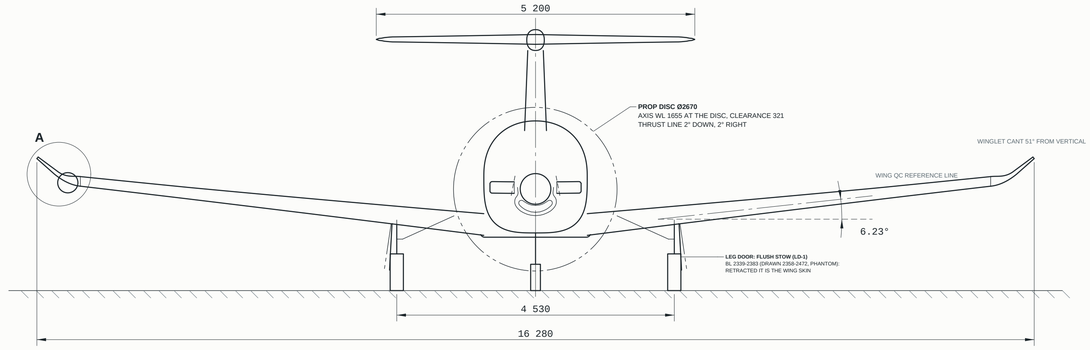


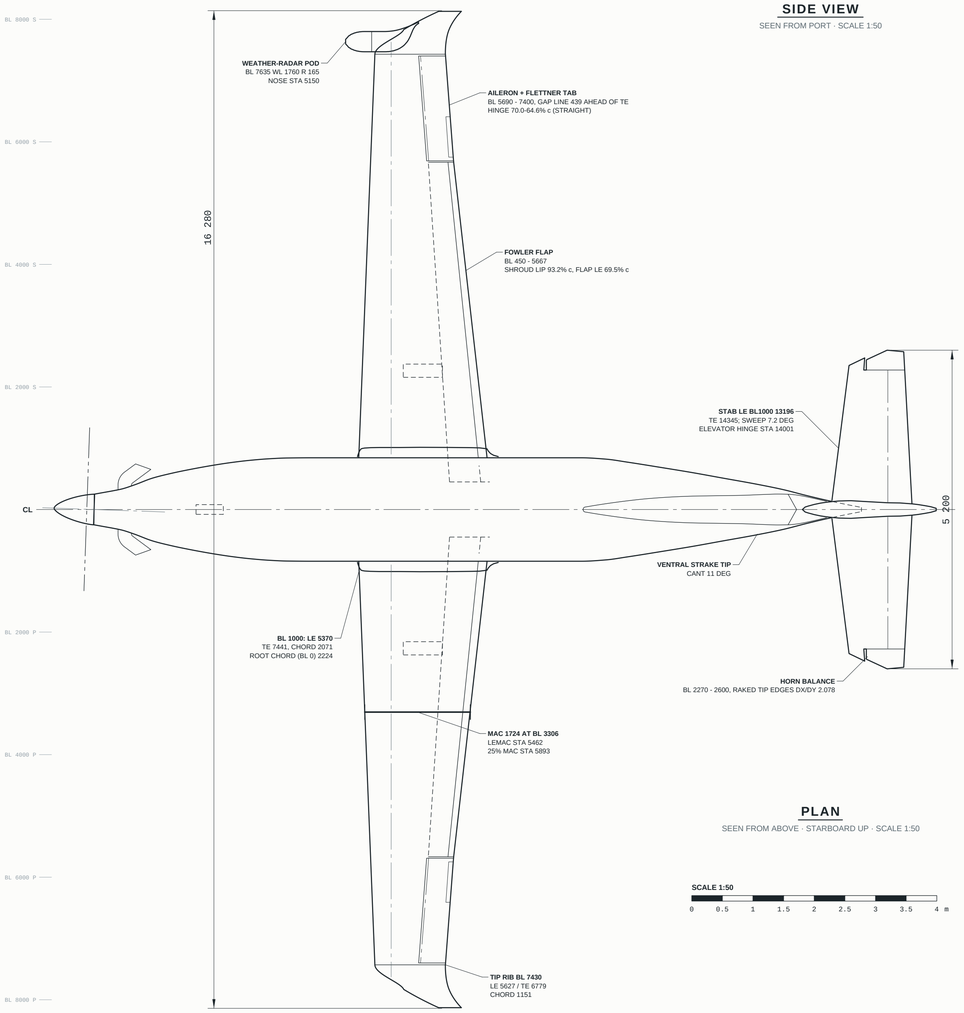
SIDE VIEW

SEEN FROM PORT · SCALE 1:50



FRONT VIEW

SEEN FROM AHEAD · STARBOARD LEFT · SCALE 1:50



PLAN

SEEN FROM ABOVE · STARBOARD UP · SCALE 1:50

SCALE 1:50

0 0.5 1 1.5 2 2.5 3 3.5 4 m

DEVIATIONS FROM THE PILATUS DRAWING (mm / deg; D = ours - dwg)

ITEM	OURS	D DWG	REV A D DWG	NOTE
Wing LE, BL 1000 (plan)	5 370	0	-104	straight LE x = 5.3302 + 0.040 y
Wing TE, BL 1000 (plan)	7 441	-12	-166	root chord solved for 25.81 m2 total plan area
Wing LE, BL 4000	5 490	0	-149	
Wing TE, BL 4000	7 102	-9	-138	
Wing LE, BL 7400 (tip rib 7430)	5 626	0	-166	
Wing TE, BL 7400 (aileron)	6 781	-7	-169	TE kink at BL 5680 (constant-chord aileron)
Flap shroud lip, BL 2000 (plan)	7 197	-5	-	FLAP_X_LIP 93.15 % (was 74.5 % · 363)
Flap shroud lip, BL 5000	6 889	+5	-	flap LE (hidden) 69.5 % unchanged
Aileron gap line, BL 5750 (plan)	6 468	-8	-	AIL_GAP_TE 439 ahead of the TE (was 68.5 % c)
Aileron gap line, BL 7395	6 342	-6	-	straight hinge line; our TE is 7 fwd
Wing upper WL, BL 4000 (front)	1 530	-1	-166	QC WL 1023 + 6.23 deg dihedral refit (rev B airfoils)
Wing lower WL, BL 4000 (front)	1 293	+2	-179	
Wing upper WL, BL 7000 (front)	1 817	0	-259	
Wing lower WL, BL 7000 (front)	1 653	-1	-269	
Dihedral (front, mean line) deg	6.10	-0.05	-1.65	chord plane 6.23; WR dim line reads 5.0
Incidence WR1 deg	1.05	0.00	+0.59	twist table (BL 0 / 5560 / tip)
Incidence WR2 deg	-0.91	0.00	+0.79	
Incidence WR3 deg	-1.93	-0.05	+1.34	
Incidence WR4 deg	-2.61	0.00	+1.79	
Section WR1 outline, max / rmss (shape)	t/c 16.5 %	4 / 2	51 / 27	CST airfoils fitted to WR1-4 (LE / TE normalised)
Section WR2 outline, max / rmss (shape)	t/c 13.8 %	4 / 1	21 / 12	rev A: modified 4-digit, too full aft of 45 % c
Section WR3 outline, max / rmss (shape)	t/c 13.2 %	3 / 1	18 / 10	
Section WR4 outline, max / rmss (shape)	t/c 12.8 %	3 / 1	16 / 9	
Winglet top WL (front)	2 180	+47	-273	ACCEPTED, FIXED-FACT DRIVEN: 16.28 span (note 5, owner)
Winglet outer BL (span / 2)	8 140	+65	+65	ACCEPTED, FIXED-FACT DRIVEN: drawn span 16.137
Winglet cant from vertical deg	51.00	+1.11	-19.89	
Stab LE, BL 1000 (plan)	13 196	0	+413	LE sweep 7.2 deg
Stab TE, BL 1000 (plan)	14 345	0	+275	
Stab LE, BL 2270 (tip rib)	13 357	0	+574	
Stab TE, BL 2270	14 277	0	+396	
Stab fixed-tip LE, BL 2400	13 470	0	-	raked edges dx/dy 2.078 (was quadratic: -30)
Stab horn LE, BL 2520	13 823	0	-	horn slot at STA 13610-13650 (was -111)
Stab tip edge, BL 2580	14 206	0	-	raked tip to the TE corner BL 2575 (was +54)
Elevator hinge STA (HF1)	14 001	0	+316	70 % chord = constant STA 14000
Stab chord plane WL (front)	4 099	0	+26	
Bullet nose STA	12 606	0	+344	
Bullet end STA (aft-most)	14 790	-21	-21	14.40 m overall length kept (drawn 14.421)
Fin LE STA at WL 3400 (side)	12 538	0	+148	LE 40.3 deg
Fin LE STA at WL 3800	12 877	0	+128	
Rudder TE STA at WL 2609 (VF1)	13 852	0	+626	TE 22.2 deg
Rudder TE STA at WL 3919 (VF2)	14 386	-3	+288	
Rudder hinge STA, WL 2609	13 251	+1	+356	nose-circle centre of the drawn rudder
Rudder hinge STA, WL 3919	13 959	0	+150	
Rudder bottom edge WL, STA 13000 / 13400	1 818 / 1 884	+1 / 0	-	sloped, on the ventral edge (rev B.0 flat 1800: -17 / -84)
Rudder top edge WL, STA 13940 / 14200	3 802 / 3 782	0 / 0	-	sloped RUD_TOP_EDGE (rev B.0 flat 3820: +18 / +36)
Fin tic at VF2 (%)	18.0	+0.2	-7.6	NACA 0018
Dorsal stab half-width FR38 / FR40	110 / 114	-2 / 0	-	straight top edge from FR33, 0.134 m/m
Dorsal root fillet hw (plan), STA 10000 / 11000 / 12200	170 / 222 / 252	0 / 0 / 0	-	E.DORSAL_FILLET foot (rev B: slab only 87 / 112 / 114)
Strake tip FR38 (BL / WL)	536 / 1 430	-13 / +1	not on FR38/40	rev A strakes STA 12350-13750
Strake tip FR40 (BL / WL)	401 / 1 528	-24 / 0	not on FR38/40	rev A strakes STA 12350-13750
Nose axle STA	2 930	+17	+37	wheelbase 3480 (PCH) centred on the drawn axles
Main axle STA	6 410	-18	+2	
Main / nose tyre OD (side)	640 / 445	+81 / 0	-	8.50-10 Type III (makers, photos): owner decision 2026-09-26
Main / nose axle WL = loaded tyre R	279 / 207	0 / -16	-	static deflection 41 / 15.5 (photos); drawn circles on the ground
Track (front)	4 530	+5	+5	Pilatus 4530
Radar pod nose STA (plan)	5 150	+2	-270	starboard wing tip; rev A at BL 3900
Radar pod axis BL / WL (front)	7 635 / 1 760	-4 / 0	-373 / -544	
Radar pod radius	165	+11	+21	PRO: enlarged for the 12-in GWX 8000
Exhaust stack outer BL / centre WL (front)	744 / 1 683	-3 / -11	-	STACK_ROOT / STACK_BEND (VQA r2); drawn nose / axis 15-18 high
Exhaust outlet, aft (inner) lip STA (plan)	1 971	+6	-	outlet faces aft-outboard; PP-STACK_SCARF (rev B plain end 1844)
Chin inlet mouth BL tip / WL bottom (front)	274 / 1 278	0 / 0	-	PP.CHIN_INLET fitted (rev B mouth 225 / 1212)
Prop disc centre STA (side)	925	0	-125	PP.PROP_X on the thrust axis (hub)
Prop disc tilt (top fwd) / yaw (stbd aft) deg	2.00 / 2.00	0.00 / 0.00	-2.00 / -2.00	THRUST_TILT_DEG / _YAW_DEG: 2 deg nose-down, 2 deg right
Prop clearance (side)	321	-16	-17	Pilatus 320 kept (+0.8 with the tilt)
Spinner outline, side (rms / max)	-	13.2 / 15.7	-	drawn axis 15 higher (WL 1655 kept); untilted 15 / 20
Spinner outline, plan (rms / max)	-	1.4 / 3.0	-	profile refit on the axis; untilted 15 / 33
Fairing flat bottom WL (front)	872	0	-137	details.BELLY_FAIRING_* (single table)
Fairing bottom WL, STA 6000 (side)	873	0	-134	hidden 6465-7450 behind the gear
Fairing nose / tail end STA (side)	5 153 / 8 585	0 / 0	-233 / -285	tail lobe on the fuselage side
Fairing root fillet BL, STA 6300 (plan)	1 014	-1	-	upper fillet over the wing root
Leg door plane BL at WL 700 (front)	2 364	-51	-	LD-1 flush stow: door = wing skin retracted
Leg door lowest WL (front)	318	-20	-	= side-view tip WL 318 (the drawn views differ by 20)
Leg door tip (side) STA / WL	6 044 / 318	-17 / 0	-	unit shifted -17.5 (wheelbase 3480)
Leg door face outline, unit frame (max / rms)	-	5 / 1	-	gear.LEG_DOOR fitted to the drawn face (rev B.0: 116 / 58)
Leg door face outline, direct (max / rms)	-	21 / 12	-	incl. the -17.5 unit shift
Leg door as built (scallop, tab), unit (max / rms)	-	146 / 34	-	LD-1 tyre scallop R 332 + tab

D DWG vs the registered drawing 190.10.40.432 (sheet 1; x +/-21, z +/-2); REV A = the model before this refit.

LAYOUT PARAMETERS (model/*.py)

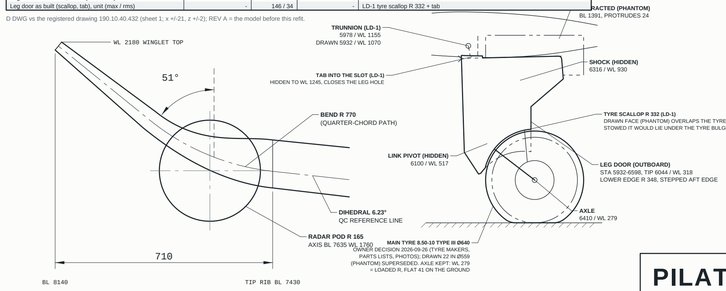
LAYOUT PARAMETER	VALUE (mm)
WING span over winglets / plan area	16 280 / 25.81 m2 (total projected)
root chord BL 0 / tip rib BL 7430	2224 / 1151 (1087 str. TE)
LE sweep / taper (straight TE)	2.29 deg / 0.489
MAC / its BL / LEMAC / 25% MAC	1724 / 3306 / 5462 / 5893
PCH aft CG 6107 in % MAC	37.4 % (rev A assumed 46 %)
dihedral / flat centre to BL	6.23 deg / 700
QC WL at BL 0 / incidence root / tip	1023 / +1.45 / -2.64 deg
sections root / tip (CST, WR1-4 fit)	LS(1)-0417MOD 16.7 % / LS(1)-0313 12.8 %
flap / aileron BL	450-5667 / 5690-7400
flap LE / shroud lip; aileron gap	69.5 / 93.15 % c; 439 ahead of TE
winglet cant / bend R / top chord	51 deg / 770 / 370
FIN · LE / rudder TE sweep	40.3 / 22.2 deg
section / rudder hinge / tab	NACA 0018 / 69.7% c / WL 2550-3600
STAB LE / TE at BL 0; chord plane WL	13070 / 14400, 4099
tip rib / horn / elevator hinge	BL 2270 / 2270-2600 / STA 14000
tip rake dx/dy / TE corner BL	2.078 / 2575
bullet / aft-most	STA 12606 - 14790
GEAR nose / main axle, wheelbase	2930 / 6410, 3 480
main / nose tyre, OD / axle WL	8.50-10 Type III 640 / 279; 17.5x6.25-6 445 / 207
main trunnion / link pivot (STA, WL)	5978,1155 / 6100,517 (drawn top 5932,1070)
retraction / retracted wheel BL / tyre	86 deg in / 1391 / 24 proud (POH ~25)
leg door STA / tip WL / BL (drawn)	5932-6598 / 318 / 2339-2383 (2358-2472)
FAIR wing-body fairing STA / bottom WL	5153-8585 / 872 (flat to BL 830)
POD radar pod axis BL / WL / R	7635 / 1760 / 165
PROP disc STA / thrust tilt / yaw	925 (hub BL +4) / 2.0 dn / 2.0 right

NOTES

1. Drawn from the parameters of model/wing.py, empennage.py, gear.py, bays.py, details.py and fuselage.py (section functions and knot tables), never from the mesh. STA = mm aft of the datum (3000 ahead of the firewall); WL above the static ground line; BL + starboard.
2. Positions fitted to the registered Pilatus NGX model drawing 190.10.40.432 (plan, side, front, sections WR1-4, VF1-2, HF1-2, FR38 / FR40); photos confirm the radar pod at the starboard wing tip.
3. Official values kept where the drawing differs: span 16.28 (drawn 16.137), wheelbase 3.48 (drawn 3.515), prop clearance 0.32 (drawn 0.336), overall length 14.40 (drawn 14.421).
4. Wing area 25.81 m2 = total projected plan area (both halves through the fuselage, incl. the aileron kink and the winglets' plan projection): the drawing's planform gives it within 0.1 %.
5. Winglet rows: ACCEPTED FIXED-FACT DEVIATION, pending the owner's decision. The official 16.28 m span (drawn 16.137) is kept by lengthening the winglet's straight part (72 mm per side, canted 51 deg), so its top is +47 mm and its outer BL +65 mm vs the drawing. No split meets 20 mm on both: every 10 mm moved into the panel semi-span lowers the top by only ~7 mm. Alternative: panel semi-span 7502 -> winglet top -4 mm, but the tip rib / aileron end +72 mm and the root chord -15 mm (area held).

PHOTO / RENDER EVIDENCE

PHOTO / RENDER (refs/photos.json)	WHAT IT SHOWS FOR THE LAYOUT
ngx_kenia_stbd_pilatus (NGX)	radar pod on the STARBOARD WING TIP leading edge, black radome, body faired into the winglet root
pro3008_stbd34_pilatus (PRO s/n 3008)	PRO pod at the starboard tip; radome slightly larger (12-in GWX 8000) -> R 165 (drawn NGX 155)
pil_techdata_front_5000 (render)	pod at the starboard tip; LE band slopes 5.7 / 6.6 deg (mean ~6.1 = drawn wing, not the 5.0 dim line)
pil_techdata_front_5000 (render)	winglet straight part ~39 deg above horizontal (cant ~51 deg), rise ~0.33 m above the tip
ngx_kenia_stbd_pilatus crop	large single leg door on the main leg, trailing link, three flap-track canoes per wing
ngx_dfbox_port (near-broadside)	leg door face = drawn side view: pointed tip, concave lower edge, stepped aft edge (sits ~0.1 m lower on the loaded, taxiing aircraft)



DETAIL A - STARBOARD WING TIP

WINGLET + RADAR POD, SEEN FROM AHEAD · 1:10

DETAIL B - MAIN GEAR

PORT UNIT, SEEN FROM PORT · 1:20

STAGE 3 - WHAT MUST FOLLOW

1. build.py verify(): 'Wing reference area' must use wing.planform_area() (total projected area incl. the aileron kink and the winglets' plan projection = 25.81); the trapezoid formula now reads 25.08.
2. Wing moved 130-160 mm aft and up (flat centre section WL 1023 at the QC, 6.23 deg dihedral from BL 700): interior.build_structure spar frames / carry-through and the cabin floor (LI: WL 1259) vs the wing upper surface at BL 0 (WL 1256, 3 mm clearance) - check fit_check.
3. Wing-to-body fairing refit (details.BELLY_FAIRING_*; livery imports them): loft the tail lobe on the fuselage side (to STA 8585) and the upper root fillet (BL 1010) - details.belly_fairing() lofts only the lower part; the centre-section airfoil (BL 0) dips below the drawn fairing bottom by up to 16 mm at STA 6112-6716.
4. Flap: inboard end BL 450 (drawn dashed under the fuselage) needs a flap well in the fairing, or keep the 3-D flap from BL 800. The flap now stows under a 93 % shroud (flap_cove / flap_loop, FLAP_TRAVEL 0.24 c): re-check the flap-track canoes and the Fowler motion in the viewer.
5. Aileron: constant-chord (gap line 439 ahead of the TE, straight hinge W.AIL_HINGE); skins / coves / body use ail_xh(y); check the Flettner tab cut-out (0.945 c) against the moving hinge fraction.
6. Airfoils refitted (model/airfoil.py rev B: CST sections fitted to the drawn WR1-4, max thickness at 30-38 % c, thin aft third): rebuild the wing skins, flap / aileron / tab sections (cut from these airfoils), coves, boots and canoes; the wing lies 13 mm lower at the root (QC WL / dihedral refit).
7. Winglet: new sections (wing winglet_sections); nav / strobe lights belong in the winglet tip (drawn light box at the winglet top), details.py still places them at the tip rib (inside the pod on starboard).
8. Gear, leg door (decision LD-1, resolved): the door is the wing lower skin carried down by the leg (gear.leg_door_offset), so retracted it closes flush (+1..+1 mm) and the tyre protrudes 24 mm in its own well. Drawn side-view face kept, scalloped round the tyre (R 332) with a hidden tab up the slot over the leg's skin crossing; edge-on it stands at BL 2339-2383 (drawn plane 2358-2472, leaning out: not possible with a flush door). Hidden changes: trunnion 5978 / WL 1155 (drawn 5932 / 1070), retraction 86 deg, side-brace stations and split, brace B0 on a lug B0 inboard of the drawn BL 2250 (the folded lower link clears the stowed leg), no forward slot.
9. Empennage: fin NACA 0018 down to WL 1760 (the ventral part below the tail cone is the drawn ventral fin); trim the rudder, tab and fin skins to the SLOPED rudder edges (E.rudder_bottom_z / rudder_top_z / rudder_outline; E.RUD_Z is now only the hinge-line WLs of those edges); trim the fuselage tail cone at the fin / rudder, model the ventral fairing (FR40 bump); horn balance now a separate plan piece (E.stab_tip_outline: slot STA 13608-13650) - split it off the fixed tip in 3-D.
10. Propeller: thrust line tilted 2 deg nose-down and yawed 2 deg right (powerplant.THRUST_*; disc STA 925 on the axis, hub BL +4): re-align the engine, mount, inlet, spinner mesh and the cowl-front ring (the spinner base centre sits 4 mm above the fuselage X0 ring centre); build.py disc construction line still at 0.80.
11. Dorsal: loft the root fillet into the tail cone (foot E.DORSAL_FILLET / dorsal_root_line, concave flank E.dorsal_fillet_section, dorsal / fin seam E.dorsal_seam) and trim the crown under it. Exhaust stacks: tube along PP.STACK_PTS / STACK_AB cut by the scarf plane PP.STACK_SCARF (exhaust_stack_rings(scarf=True)), black collar STACK_COLLAR. Chin inlet: mouth + polished lip from PP.CHIN_INLET in the keel step (D2).
12. CLAUDE.md: update the wing facts (planform from the drawing, LEMAC 5462 / MAC 1724, dihedral 6.23 deg, airfoils = CST fits to the drawing, radar pod at the starboard wing tip, gear axles 2930.5 / 6410.5, thrust line 2 deg down / 2 deg right).

PILATUS PC-12 PRO

GENERAL ARRANGEMENT - LAYOUT (WING, TAIL, GEAR)

FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L4	A	AS SHOWN	A1 · 1 OF 1	mm

DRAWN	DATE
Claude - loftkit parametric model	2026-09-24

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION